Submitted to:
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Password Manager System

COMP2004 **- DATABASE MANAGEMENT SYSTEMS**

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System Description

- **The Problem:** Modern users manage dozens of digital credentials, risking data breaches and poor password hygiene.

- **Our Solution:** A highly structured, relational database tailored for identity isolation, session auditing, and cryptographic secret tracking.

- **Creative Core:** Moving beyond standard business systems to build a highly secure ecosystem incorporating multi-device logging and hardware verification.

Functional Requirements

User Identity Isolation:

Registration via globally unique, verified electronic mail channels.

01

Taxonomy & Separation:

Decoupling platform domains (Websites) from user entries (Accounts) grouped by administrative categories.

02

Cryptographic Isolation:

Enforcing a strict 1:1 mapping between account profiles and encrypted data layers.

03

Security Auditing:

Tracking live hardware device fingerprints, transactional session logs, and access IP locations.

04

Logical Design (E-R Diagram)

Star Layout Optimization

Structured for transactional lookup speed and referential mapping safety.

Strict 1:1 Bound

Enforced physically by placing a UNIQUE constraint directly on Passwords.accountID for advanced cryptographic segregation.

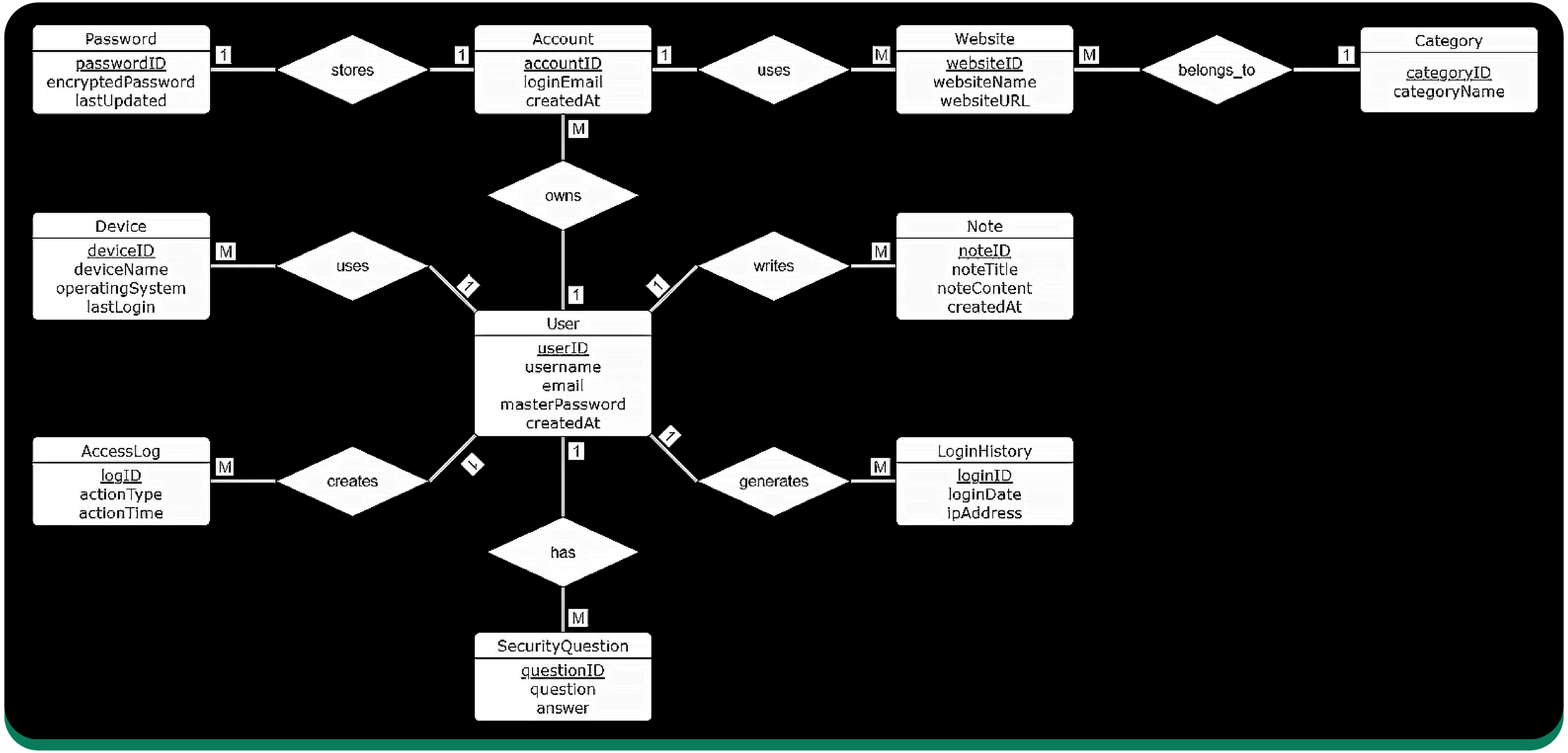
Bridge Resolution (M:1)

The Accounts entity resolves the many-to-many relationship between Users and Websites.

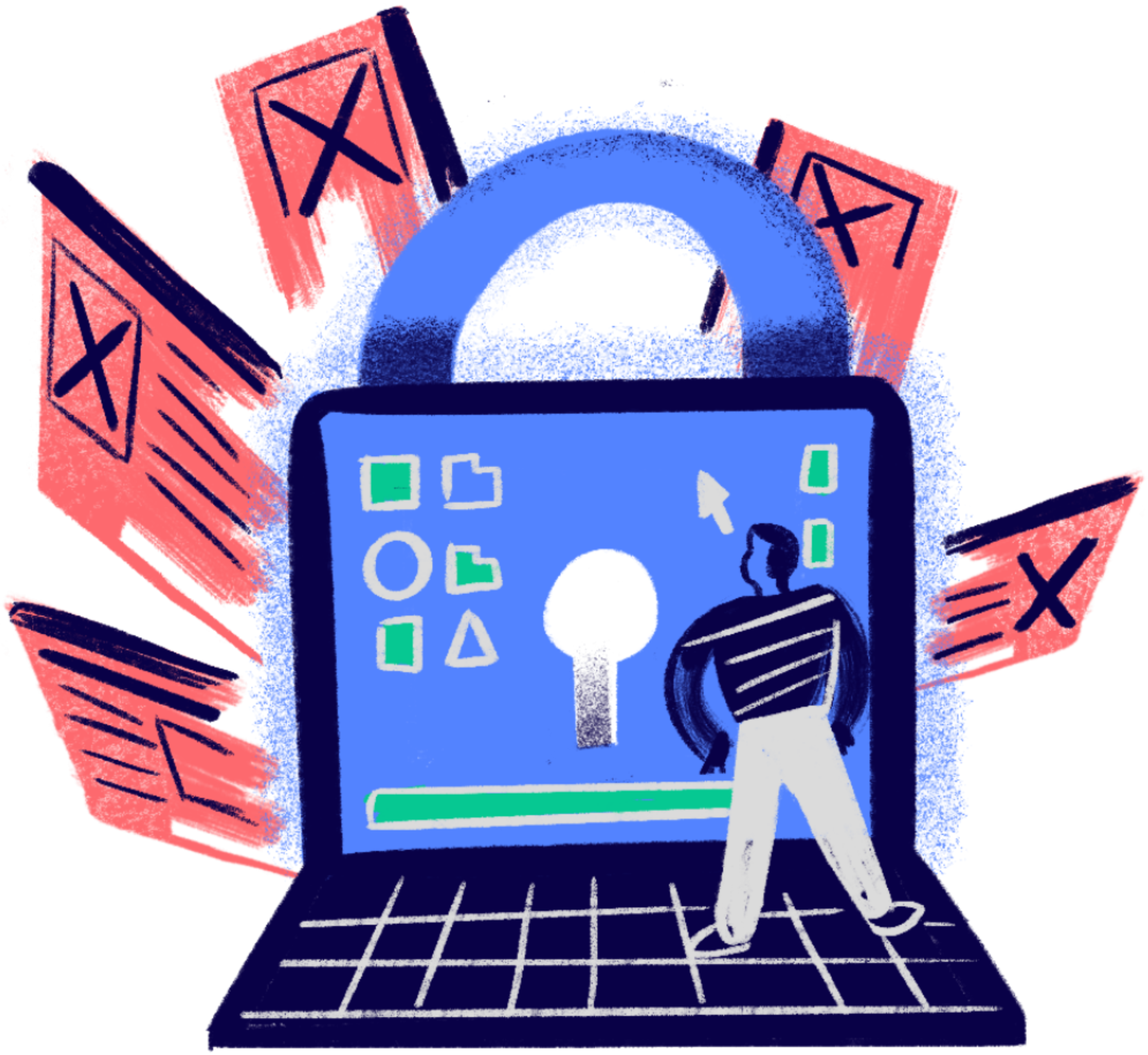
Composite Unique Constraint

Locked down via UQ_Account (userID, websiteID, loginEmail) to eliminate duplicate vault rows.

Logical Design (E-R Diagram)



Physical Schema & Constraints (3NF) 🔗



Normalization Standard:

Normalization Standard: 10 fully related tables meeting strict Third Normal Form (3NF) criteria (Exceeding the 8-table minimum requirement).

Data Packing Optimization:

Scaled native INT fields for keys, and VARCHAR(255) allocation for high-entropy base64 encrypted blocks.

Strict Integrity Constraints:

UNIQUE NOT NULL constraints applied to core identifiers like Users.email to preserve credential uniqueness.

Domain boundaries enforced via CHK_ActionType in AccessLogs to validate live transactional audit trails.

Data Volume Dashboard 🗂

⚠ Total System Volume 105 Records ⚠

25	USERS CATEGORIES WEBSITES	PRIMARY KEY UNIQUE Email NOT NULL
30	ACCOUNTS PASSWORDS	FOREIGN KEY UNIQUE 1:1 Constraint Composite UQ
30	NOTES DEVICES SEC.QUE.	FOREIGN KEY to Users NOT NULL Title
20	LOGINHISTORY ACCESSLOGS	DATETIME precision CHK_ActionType

SQL Implementation Hierarchy

01

DML Execution Suite (12 Operations):

Simulating new user onboarding, administrative routing changes (Twitter to X via pattern matching), and secure data purging.

02

Simple Queries (7 Operations):

Covering single-table filtration, string scanning over user notes, and system log isolation.

03

Advanced Relational Diagnostics (5 Operations):

Complex business logic using Multi-table Joins, Aggregations (GROUP BY / HAVING), CTEs (WITH clauses), and Server VIEWS.

Live Demo & Verification ((•))

Environment:

- Microsoft SQL Server (T-SQL) via SSMS.

Live Scenarios to Demonstrate:

- Structural Schema Building & Referentially Safe Seed Feeding.
- Advanced Threat Detection Auditing (Catching abnormal frequencies of password lookups).
- Relational Set Operations via HighRiskBankingProfiles View.

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Thank You!

We welcome any questions or requests for further clarification.

CONTACT

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